

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2020 – 2021 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. EEE

Course Code & Title: EE8691 Embedded Systems

Name of the Faculty member: S. Sharmila Kumari

ACTIVITY

Topic: Watch Dog Timer, Real Time Clock, In Circuit Emulator

Goal (Learning Outcome):

- ❖ The students will be able to compare Watch Dog Timer, Real Time Clock and In Circuit Emulator – (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Mini - map

Justification for Choosing the Activity for the topic:

1. Differentiate Watch Dog Timer, Real Time Clock and In Circuit Emulator

After teaching the concept, I thought of conducting this activity for making the students to give the difference between the three concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, the students were made to pair with their neighbors

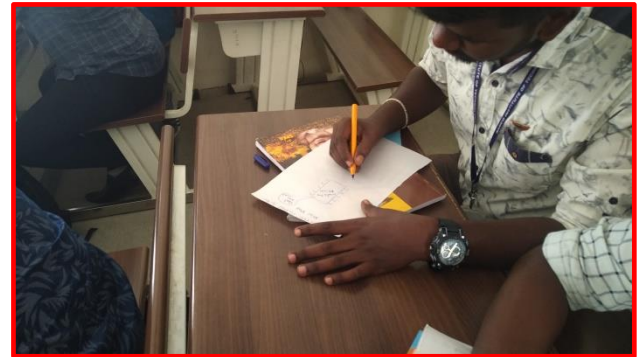
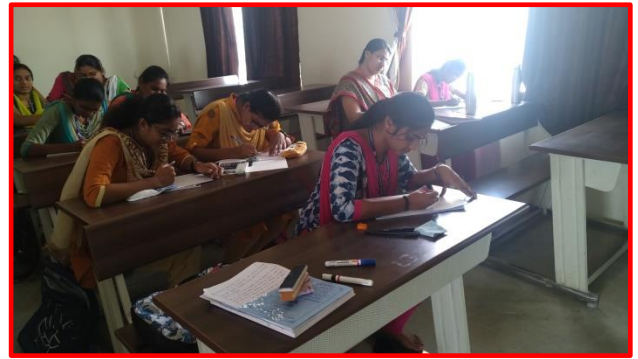
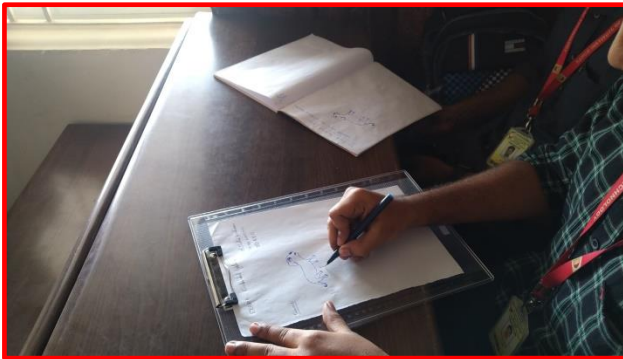
Photographer: Myself

Reporter: Myself

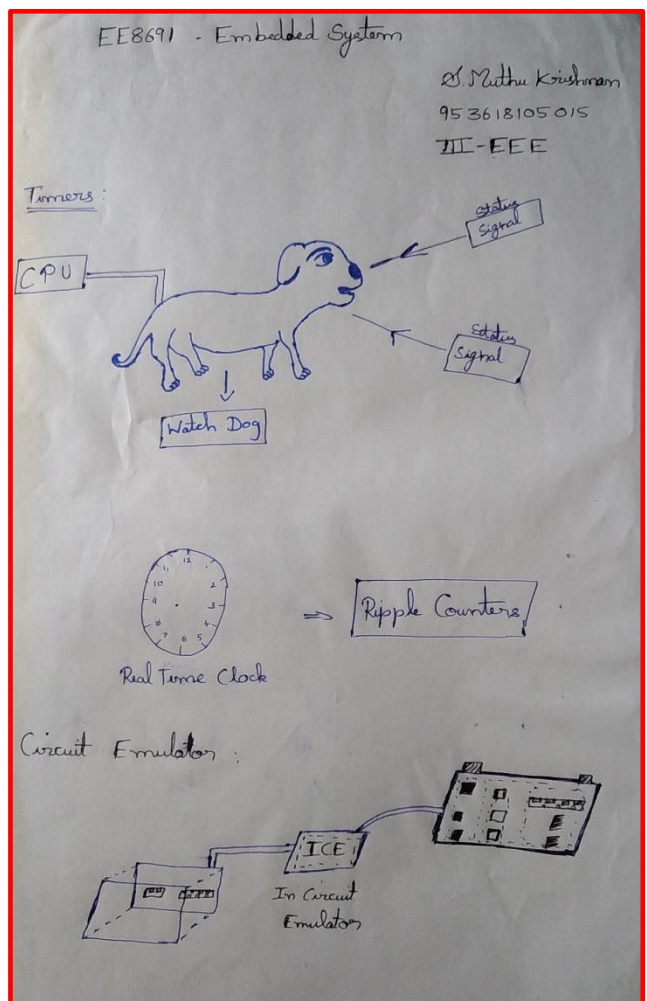
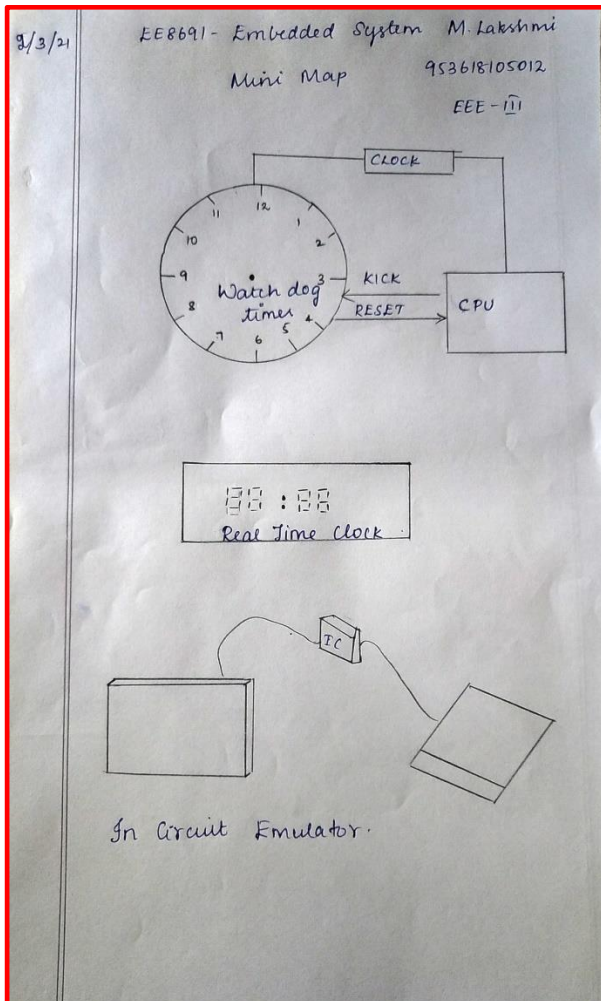
At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about Watch Dog Timer, Real Time Clock and In Circuit Emulator concept for 2 minute.
- ✓ Then I told them to Pair with their neighbours and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design mini-map and submit within 3 minutes.

A view of implementation of the Activity



Samples:



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain the concepts of Watch Dog Timer, Real Time Clock and In Circuit Emulator.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 50% of students answered.**

Reflective Critique:

1. Pre-implementation Reflection :

I preferred this Activity because they have Watch Dog Timer, Real Time Clock and In Circuit Emulator separately.

Challenges anticipated:

- ❖ In the class mostly boys hastate to answer to the questions.
- ❖ Time utilization for conducting activity.

Steps taken:

- ❖ While using this activity, the students will get clarity in the timer and counter concept.
- ❖ The boys are sitting in 2 columns – I planned to choose more pairs from boys to involve them in the activity.

2. Post implementation Reflection:

• Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• Challenges:

- Slow learners were not able to understand some topics during discussion hours.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C311.1	2	2	2	1	1	2	-	1	-	1	-	-

References:

1. <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-mini-maps>

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ACTIVITY

Topic: Serial Bus Communication Protocol

Goal (Learning Outcome):

- ❖ The students will be able to compare RS 232, RS 422 and RS 485 – (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Visual Clicker

Justification for Choosing the Activity for the topic:

2. Differentiate RS 232, RS 422 and RS 485

The topic Serial Bus Communication Protocols (RS 232, RS 422 and RS 485) are used for the serial communication purpose. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the three concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept of RS 232, RS 422 and RS 485, the students were made to pair with their neighbors

Total Strength is 55, Number of Pairs – minimum 6 in one team

Photographer: one student – Mr. V. R. Ugesh (interested in photography)

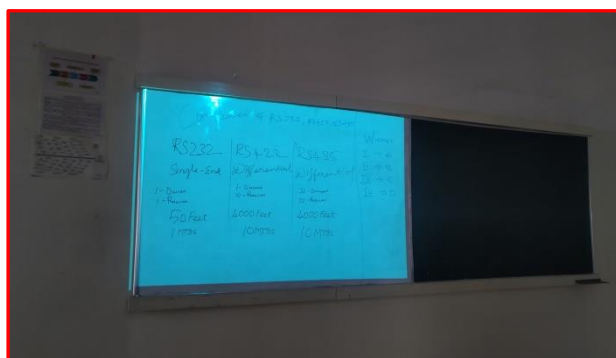
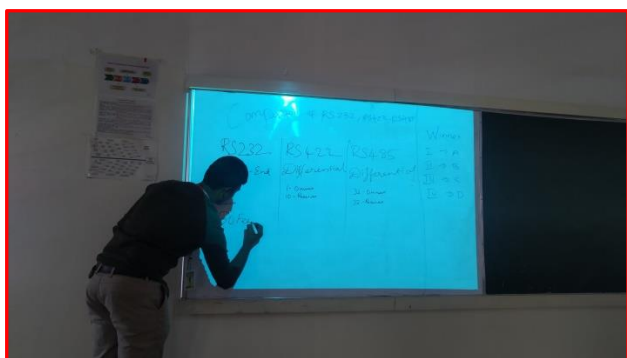
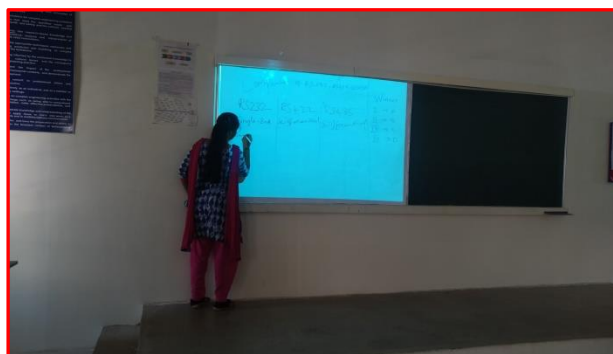
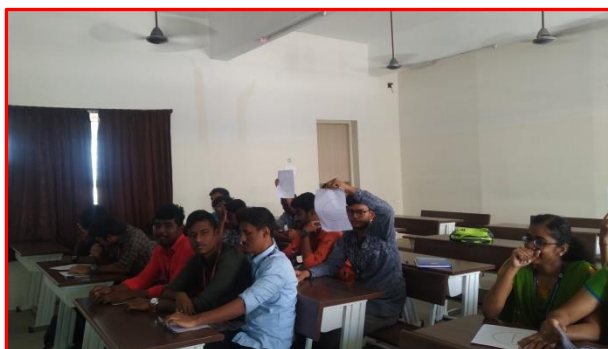
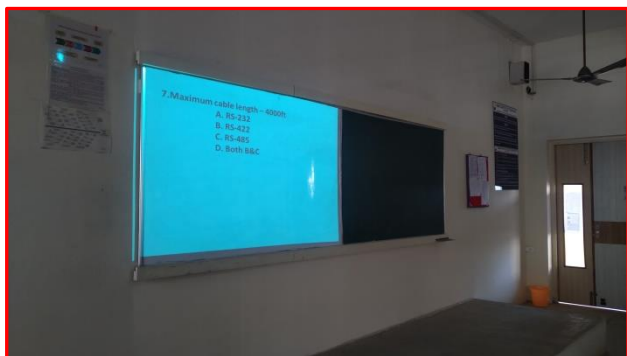
Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about RS 232, RS 422 and RS 485 concept for 1 minute.
- ✓ Then I told them to Pair with their neighbors and discuss for another 1 minute.

- ✓ With the help of PPT prepared (Multiple choice questions) students use their clickers to respond to the question and then the answers were be projected to the students.
- ✓ After completion of the class I uploaded the PPT in the course website (Canvas)

A view of implementation of the Activity



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain the concepts of RS232, RS422 and RS485.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 50% of students answered.**

Reflective Critique:

3. Pre-implementation Reflection :

I preferred this Activity because they have RS232, RS422 and RS485 separately.

Challenges anticipated:

- ❖ Time utilization for conducting activity.

4. Post implementation Reflection:

- **Benefits:**

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- **Challenges:**

- Student clicker's were not clear in photos. So clicker tools prepared from students were not advisable.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C311.2	2	3	2	1	1	2	-	1	-	1	-	1

References:

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